

CS432: Databases

Introduction to the Relational Model

Instructor
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Lecture no.
5

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Structure of Relational Databases

A relational database consists of a collection of tables

<i>course_id</i>	<i>title</i>	<i>dept_name</i>	<i>credits</i>
BIO-101	Intro. to Biology	Biology	4
BIO-301	Genetics	Biology	4
BIO-399	Computational Biology	Biology	3
CS-101	Intro. to Computer Science	Comp. Sci.	4
CS-190	Game Design	Comp. Sci.	4
CS-315	Robotics	Comp. Sci.	3
CS-319	Image Processing	Comp. Sci.	3
CS-347	Database System Concepts	Comp. Sci.	3
EE-181	Intro. to Digital Systems	Elec. Eng.	3
FIN-201	Investment Banking	Finance	3
HIS-351	World History	History	3
MU-199	Music Video Production	Music	3
PHY-101	Physical Principles	Physics	4

Course table

Structure of Relational Databases

A relational database consists of a collection of tables

<i>course_id</i>	<i>prereq_id</i>
BIO-301	BIO-101
BIO-399	BIO-101
CS-190	CS-101
CS-315	CS-101
CS-319	CS-101
CS-347	CS-101
EE-181	PHY-101

prereq table

Relations

- **prereq**: Two courses are related
- **instructor**: Relationship between a specified ID and the corresponding values for name, dept_name, and salary values.
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A row in a table represents a relationship among a set of values.

A table is a collection of such relationships.

Mathematical terminology to Relational

- A tuple is simply a sequence (or list) of values
- A relationship between n values is represented mathematically by an n -tuple of values.
- This corresponds to a row in a table.
- A relation is a set of Tuples.
- An attribute refers to a column of a table.

Ordering does not matter!

<i>ID</i>	<i>name</i>	<i>dept_name</i>	<i>salary</i>
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
98345	Kim	Elec. Eng.	80000
76766	Crick	Biology	72000
10101	Srinivasan	Comp. Sci.	65000
58583	Califieri	History	62000
83821	Brandt	Comp. Sci.	92000
15151	Mozart	Music	40000
33456	Gold	Physics	87000
76543	Singh	Finance	80000

Relation instance: Specific instance of a relation. Above instance has 12 rows/tuples.

Attribute domain

A set of permitted values for a given attribute.

- **salary attribute:** Six figure salary.
- **name attribute:** Only English alphabets.
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- **Null value** is a special value that signifies that the value is unknown or does not exist.

A domain is **atomic** if elements of the domain are considered to be **indivisible** units. **Any example?**

Database Schema

- Database schema – is the logical structure of the database.
Instructor (ID, name, deptname, salary)
- Database instance – is a snapshot of the data in the database at a given instant in time.

<i>ID</i>	<i>name</i>	<i>dept_name</i>	<i>salary</i>
10101	Srinivasan	Comp. Sci.	65000
12121	Wu	Finance	90000
15151	Mozart	Music	40000
22222	Einstein	Physics	95000
32343	El Said	History	60000
33456	Gold	Physics	87000
45565	Katz	Comp. Sci.	75000
58583	Califieri	History	62000
76543	Singh	Finance	80000
76766	Crick	Biology	72000
83821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000

For time-being, the precise definition of the domain of each attribute is not considered.

Keys

- Let $K \subseteq R$
- K is a **superkey** of R if values for K are sufficient to identify a unique tuple of each possible relation $r(R)$
 $\{ID\}$ and $\{ID, name\}$ are both superkeys of instructor
- Superkey K is a **candidate key** if K is minimal
 $\{ID\}$ is a candidate key for Instructor
- One of the candidate keys is selected to be the **primary key**.
Which one?
- **Foreign key constraint**: Value in one relation must appear in another
 - Referencing relation
 - Referenced relation
 - deptname in instructor is a foreign key from instructor referencing department

Keys

A way to specify how tuples within a given relation are distinguished

- **Superkey:** A set of one or more attributes that allow us to identify uniquely a tuple in the relation.

<i>ID</i>	<i>name</i>	<i>dept_name</i>	<i>salary</i>
10101	Srinivasan	Comp. Sci.	65000
12121	Wu	Finance	90000
15151	Mozart	Music	40000
22222	Einstein	Physics	95000
32343	El Said	History	60000
33456	Gold	Physics	87000
45565	Katz	Comp. Sci.	75000
58583	Califieri	History	62000
76543	Singh	Finance	80000
76766	Crick	Biology	72000
83821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000

Possible superkeys: $\{ID\}$, $\{ID, name\}$, ...

Superkey

More formally:

- A = set of attributes in the schema of relation r .
- $K \subseteq A$ is a superkey for r .
- Given any two tuples t_1 and t_2 from r and $t_1 \neq t_2$, then:

$$t_1.K \neq t_2.K$$

Superkey

More formally:

- A = set of attributes in the schema of relation r .
- $K \subseteq A$ is a superkey for r .
- Given any two tuples t_1 and t_2 from r and $t_1 \neq t_2$, then:

$$t_1.K \neq t_2.K$$

No two distinct tuples have the same values on all attributes in K .

Keys

A way to specify how tuples within a given relation are distinguished

- **Candidate Keys:** Superkeys for which no proper subset is a superkey.
 - $\{ID, name\}$ ✗
 - $\{ID\}, \{name, dept_name\}$ ✓

Minimal superkeys are called candidate keys

Keys

A way to specify how tuples within a given relation are distinguished

- **Primary key:** A candidate key that is chosen by the database designer as the principal means of identifying tuples within a relation
 - Never, or very rarely, changed
 - May have geographical dependence
 - Mostly listed first in the schema
 - Primary key attributes are also underlined

Keys

A way to specify how tuples within a given relation are distinguished

- **Foreign key:** A relation, say r_1 , may include among its attributes the primary key of another relation, say r_2 . This attribute is Foreign Key.
 - r_1 : Referencing relation
 - r_2 : Referenced relation

Any examples from previous tables?

Exercise: Keys

— Let us consider the student database with tables for Students, Courses, and Enrollment and draw their associated keys.

- students

<u>roll#</u>	first_name	last_name	DOB	passport#	aadhaar#	department
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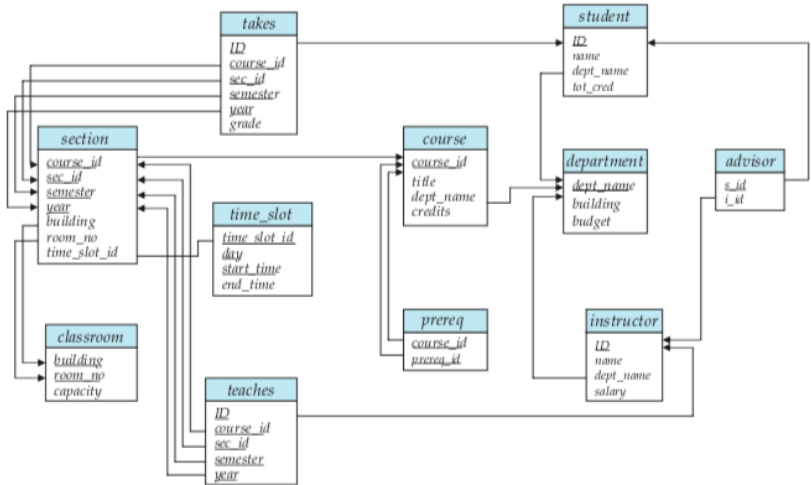
- courses

<u>course#</u>	course_name	credits	LTP	department
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- enrollment

<u>roll#</u>	<u>course#</u>	instructorID
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Schema Diagrams



Relational Operations

- Selection specific tuples

<i>ID</i>	<i>name</i>	<i>dept_name</i>	<i>salary</i>
12121	Wu	Finance	90000
22222	Einstein	Physics	95000
33456	Gold	Physics	87000
83821	Brandt	Comp. Sci.	92000

Result of query selecting instructor tuples with salary greater than \$85000

Relational Operations

- Select certain attributes

<i>ID</i>	<i>salary</i>
10101	65000
12121	90000
15151	40000
22222	95000
32343	60000
33456	87000
45565	75000
58583	62000
76543	80000
76766	72000
83821	92000
98345	80000

Result of query selecting attributes 'ID' and 'salary' from the instructor relation.

Relational Operations

- Join Operation

<i>ID</i>	<i>name</i>	<i>salary</i>	<i>dept_name</i>	<i>building</i>	<i>budget</i>
10101	Srinivasan	65000	Comp. Sci.	Taylor	100000
12121	Wu	90000	Finance	Painter	120000
15151	Mozart	40000	Music	Packard	80000
22222	Einstein	95000	Physics	Watson	70000
32343	El Said	60000	History	Painter	50000
33456	Gold	87000	Physics	Watson	70000
45565	Katz	75000	Comp. Sci.	Taylor	100000
58583	Califieri	62000	History	Painter	50000
76543	Singh	80000	Finance	Painter	120000
76766	Crick	72000	Biology	Watson	90000
83821	Brandt	92000	Comp. Sci.	Taylor	100000
98345	Kim	80000	Elec. Eng.	Taylor	85000

Result of natural join of the instructor and department relations.

Relational Operations

- Cartesian product
- Union
- Intersection
- Set difference

Relational algebra will cover in detail in later lectures!

Acknowledgments/Contributions

- Some of the images utilized in these slides are subject to copyright - Abraham Silberschatz, Henry Korth, and S. Sudarshan. Database System Concepts. 6th Edition, McGraw-Hill Education
- Contributor 2024-25, 2025-26 - Yogesh K. Meena, IIT Gandhinagar